

CHARBEL HABER

Mid-to-Senior Software Engineer

+961 71 04 76 45 | charbel.el.haber@gmail.com | Mrouj - Maten, Lebanon | github.com/charbelhaber

PROFESSIONAL SUMMARY

Performance-driven Mid-to-Senior Software Engineer with nearly 4 years of professional experience specializing in secure backend architectures, real-time IoT telemetry gateways, and scalable web applications. Proven track record in implementing advanced backend concepts including robust RESTful API lifecycles, authentication/authorization layers, rate limiting, and traffic throttling. Expert in designing low-latency communication systems, containerized environments, and critical notification engines using Twilio while leveraging AI-assisted coding methodologies to maximize lifecycle development efficiency.

CORE SKILLS

Backend Architecture:	Node.js, Python (FastAPI, Django), PHP (Laravel), RESTful APIs (GET, POST, PUT, DELETE)
Advanced Backend:	Authentication (JWT, OAuth2), Authorization (RBAC), Rate Limiting, Throttling, Pagination
Real-Time & Notifications:	WebSockets, MQTT, Twilio API (Push Notifications & SMS Gateway), CAN bus protocols
Frontend Development:	React, TypeScript, JavaScript, Tailwind CSS, Bootstrap, HTML5/CSS3
Mobile Development:	Flutter, Dart (Cross-Platform Mobile Apps)
Databases & Caching:	PostgreSQL, MySQL, Redis, Database Management (SQL), Sequelize
DevOps & AI:	Linux, Docker, Docker Compose, Git, AI-Assisted Coding, Prompt Engineering, ML (YOLOv5)

WORK EXPERIENCE

Full Stack Software Engineer	Dec 2022 – Present
Beytek SAL Offshore • Jal El Dib, Lebanon	
Central Notification Engine (Twilio): Designed and deployed a centralized communications hub utilizing the Twilio API to broadcast low-latency push notifications, emergency alerts, and system-state SMS updates directly to field engineers and operational dashboards. Secure Backend & API Lifecycle: Designed and optimized enterprise-grade RESTful APIs using FastAPI and Node.js, implementing proper HTTP verbs (GET, POST, PUT, DELETE), cursor/offset-based pagination, and strict validation layers. Advanced Gateway Security: Engineered robust application security mechanisms including stateful/stateless authentication, role-based authorization, token verification, client-based rate limiting, and server-side traffic throttling to safely expose IoT telemetry nodes. Architected Real-Time Telemetry: Designed and deployed low-latency WebSocket and MQTT servers/clients for robust cloud-to-device communication, ensuring seamless synchronization with embedded hardware (Raspberry Pi, Micrologix, Protonode). Advanced IoT Integration: Engineered and integrated industrial elevator control systems, managing real-time CAN communication protocols, state tracking, car operations, and high-frequency hall calls. Frontend Revamp: Spearheaded the overhaul of critical system monitoring and technician applications, transitioning legacy codebases into highly responsive, interactive interfaces using React, TypeScript, and Tailwind CSS. DevOps & Containerization: Streamlined local development and automated deployment pipelines using Docker and Docker Compose across both embedded Linux nodes and cloud infrastructures. Leadership & Mentorship: Actively mentored junior developers and engineering interns, conducting code reviews, aligning design principles, and fostering best practices in software architecture.	

- **Predictive Analytics:** Implemented time-series power consumption forecasting models and computer vision pipelines utilizing YOLOv5 and FasterRCNN.

IT Specialist

Jan 2022 – Nov 2022

Lebanese Alternative Learning (LAL NGO) • Beirut, Lebanon

- **Embedded System Deployments:** Managed server optimization and automated deployment of Moodle platforms onto Raspberry Pi hardware to securely host the offline Lebanese national curriculum.
- **Full-Stack Maintenance:** Administered, customized, and optimized corporate WordPress architectures and Moodle instances, ensuring high availability, speed, and responsive layout designs.
- **Cross-Functional Collaboration:** Partnered closely with educators to integrate complex digital resources, troubleshooting platform vulnerabilities and backend bottlenecks.

Developer Intern

Oct 2021 – Dec 2021

Beytek SAL Offshore • Jal El Dib, Lebanon

- Constructed responsive frontend pages consuming RESTful APIs using vanilla web technologies.
- Coordinated directly with the Senior Developer to review architectural requirements and gain deep familiarity with Node.js, Django, and Laravel.

PERSONAL PROJECTS

Qlix – AI-Powered Receipt Scanner & Excel Generator • Flutter, Dart, AI/OCR Integration

Developed and deployed a mobile application that uses OCR and Artificial Intelligence to parse physical financial receipts. Implemented automated data extraction pipelines to process raw receipt text and structurally generate formatted, downloadable Excel expense sheets for user bills.

LebPaw – Pet & Vet Ecosystem App • Flutter, Dart (In Active Development)

Architecting a comprehensive cross-platform mobile application dedicated to pet owners and veterinarians in Lebanon. Integrating location-based services and geolocation mapping to allow users to dynamically track, filter, and find the nearest vet clinics and pet stores.

EDUCATION & CONTINUING EDUCATION

Professional Certificate in AI & Prompt Engineering (Ongoing)

2026

American University of Beirut (AUB) • Beirut, Lebanon

Specializing in **AI-Assisted Coding** methodologies, prompt optimization, and integrating LLMs into software engineering workflows.

Bachelor of Science in Computer Science

Sep 2018 – Jun 2022

AI Kafaat University (AKU) • Beirut, Lebanon

LANGUAGES & INTERESTS

Languages: English (Fluent) • Arabic (Native/Fluent) • French (Fluent)

Interests: Gym & Functional Training • Tech & Gaming • Reading